5.3.1 Distance Between Two Points on the Coordinate Axes

Points on x – axis: If two points lie on the x- axis, then the distance between them is equal to the difference between the x- coordinates.

Consider two points $A\ (x_1,0)$ and $B\ (x_2,0)$ on the x-axis .

The distance of B from A is

$$AB = OB - OA = x_2 - x_1 \text{ if } x_2 > x_1 \text{ or}$$
$$= x_1 - x_2 \text{ if } x_1 > x_2$$
$$AB = |x_2 - x_1|$$

(Read as modulus or absolute value of $x_2 - x_1$)

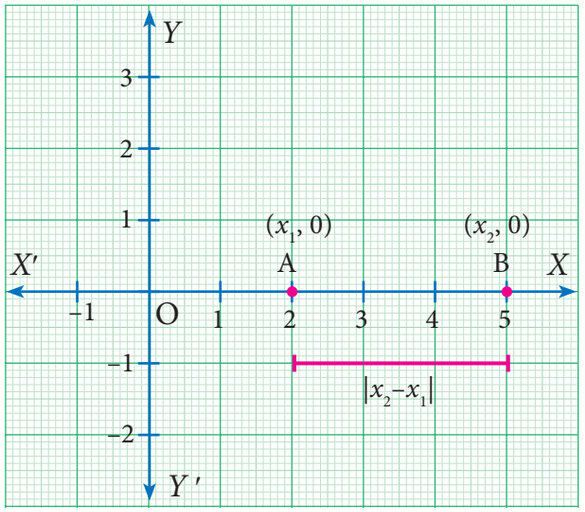


Fig. 5.14

Points on y – axis: If two points lie on y-axis then the distance between them is equal to the difference between the y-coordinates.

Consider two points $P(0,y_1)$ and $Q(0,y_2)$

The distance Q from P is

$$PQ = OQ - OP.$$
$$= y_2 - y_1 \text{ if } y_2 > y_1 \text{ or}$$
$$= y_1 - y_2 \text{ if } y_1 > y_2$$
$$PQ = |y_2 - y_1|$$

(Read as modulus or absolute value of $y_2 - y_1$)

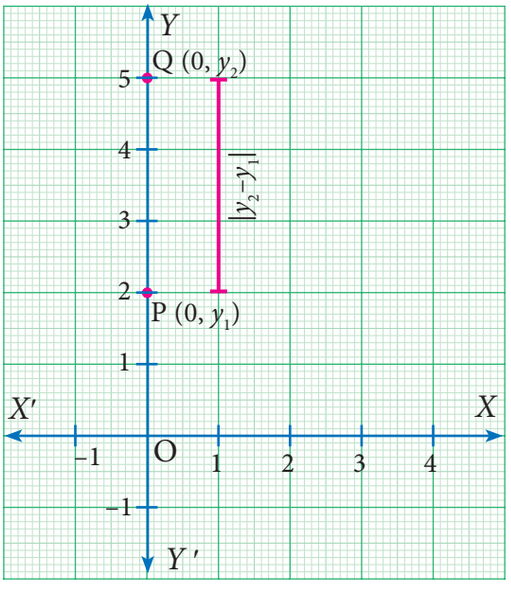


Fig. 5.15

5.3.2 Distance Between Two Points Lying on a Line Parallel to Coordinate Axes

Consider the points $A(x_1, y_1)$ and $B(x_2, y_1)$. Since the y - coordinates are equal the points lie on a line parallel to x- axis. From A and B draw AP and BQ perpendicular to x- axis respectively. Observe the given figure (Fig. 5.16), it is obvious that the distance AB is same as the distance PQ

Distance AB = Distance between $PQ = |x_2 - x_1|$
[The difference between x coordinates]
Similarly consider the line joining the two points

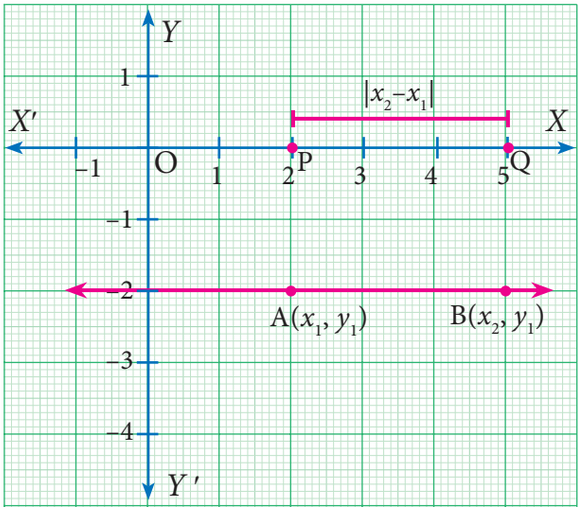


Fig. 5.16

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